Predicting Alternative Protein Conformations by Perturbing Pair Representations

Shosuke Suzuki[1] Toshiyuki Amagasa[2]

[1]Graduate School of Science and Technology, University of Tsukuba, Japan
[2]Center for Computational Sciences, University of Tsukuba, Japan

suzuki.shosuke@kde.cs.tsukuba.ac.jp
amagasa@cs.tsukuba.ac.jp

Introduction

Deep-learning models such as AlphaFold2 achieve highly accurate structure predictions, but typically output a single, static conformation, whereas many proteins are in fact dynamic molecules that must adopt multiple conformational states to carry out their biological functions. However, standard models often fail to capture this functionally important conformational diversity, and existing approaches to multi-state prediction can be computationally demanding or difficult to interpret. Here, we explore a simple and lightweight strategy to sample alternative conformations in the Boltz-2 framework by globally rescaling the model's latent pair representations, allowing us to reweight distinct conformational modes without retraining the network.

Proposed Methods: Perturbing Pair Representations

We consider Boltz-2 and study a simple strategy that we call pair-representation perturbation, which applies a single, uniform scaling factor β to the latent pair representations z_{ij}. Here, z_{ij} is a residue–residue embedding summarizing coevolutionary couplings and geometric preferences between positions i and j, and the structure module uses z_{ij} as the main interface between sequence information and 3D geometry. Scaling z_{ij} with β changes how strongly these learned couplings constrain the structure: $\beta > 0$ effectively strengthens them, whereas $\beta < 0$ attenuates them and can increase conformational diversity; concretely, we implement this as

$$z_{ij}^{\text{scaled}} = (1 + \beta)\, z_{ij}.$$

This operation has negligible computational cost, requires no retraining, and keeps the input sequence and MSA fixed, effectively reweighting structural modes already encoded in the network rather than generating new ones from scratch.

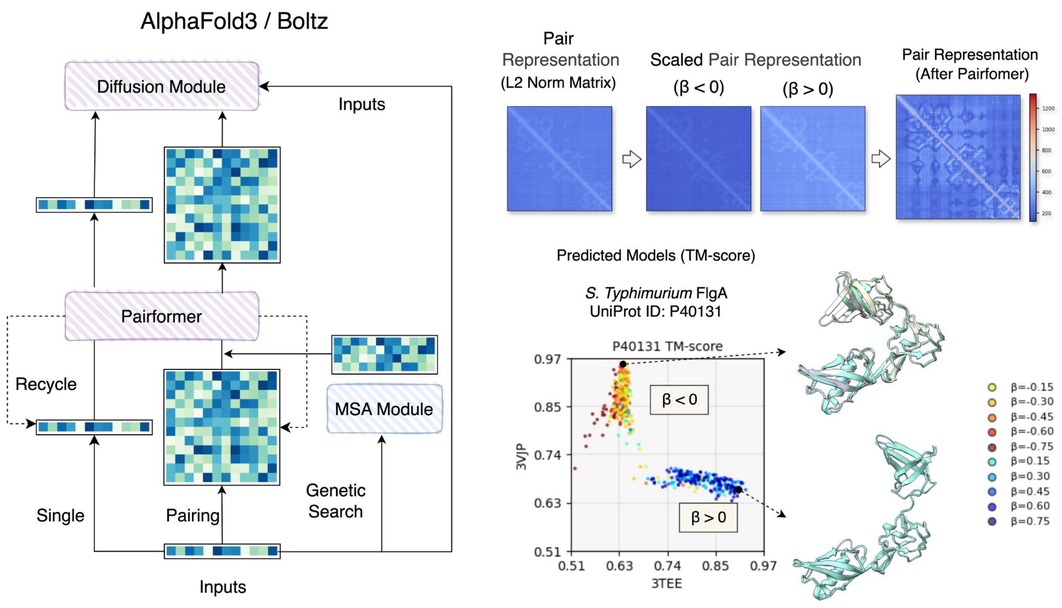

Figure 1. Overview of pair-representation perturbation. Scaling the pair representation z with β modulates sequence–structure coupling and reshapes the sampled conformational ensemble.

Results: Hinge-like Proteins & Transporters

We evaluated pair-representation perturbation on 23 hinge-like proteins (OC23) and 15 transporters (TP16 except SPF1) from the curated dataset of Kalakoti et al. (2025).

- For each target, we sampled conformations around the two reference states (open/closed for OC23, inward/outward for TP16) and computed TM-scores to both states.
- Defining an *improved* target as one where the lower of the two TM-scores increases by ≥ 0.05 relative to the default (unperturbed) model, we find that perturbing the pair representations improves the minimum TM-score for 4/23 hinge-like proteins and 7/15 transporters, with particularly strong gains on transporters.

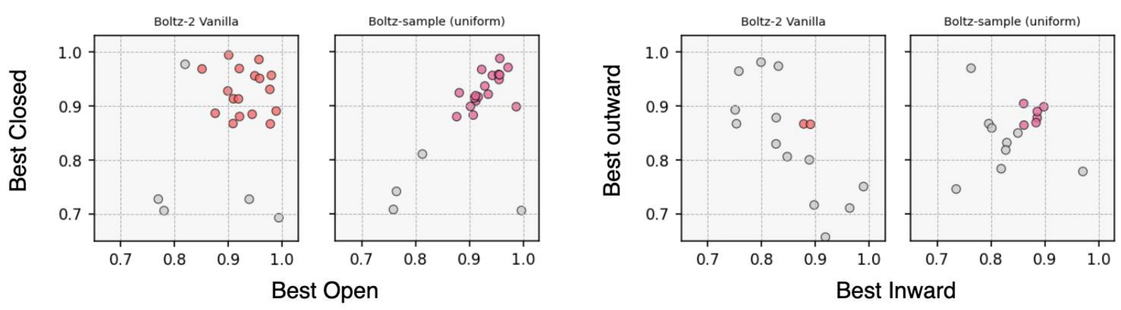

Figure 2. Best TM-scores for each target obtained with the default model and with pair-representation perturbation. The upper panels show scatter plots of the best TM-scores: best open vs. best closed for the 23 hinge-like proteins, and best inward vs. best outward for the 15 transporters. Each point corresponds to one target; red points indicate cases with TM-score ≥ 0.85.

Results: Sign-Controlled Sampling

Across multiple hinge-like proteins and transporters, changing only the sign of β can steer sampling toward one or the other of the experimental conformations for a subset of targets. Moreover, simply selecting the top 1% of models by confidence score already recovers representatives of both states, without requiring PCA or other post hoc analyses.

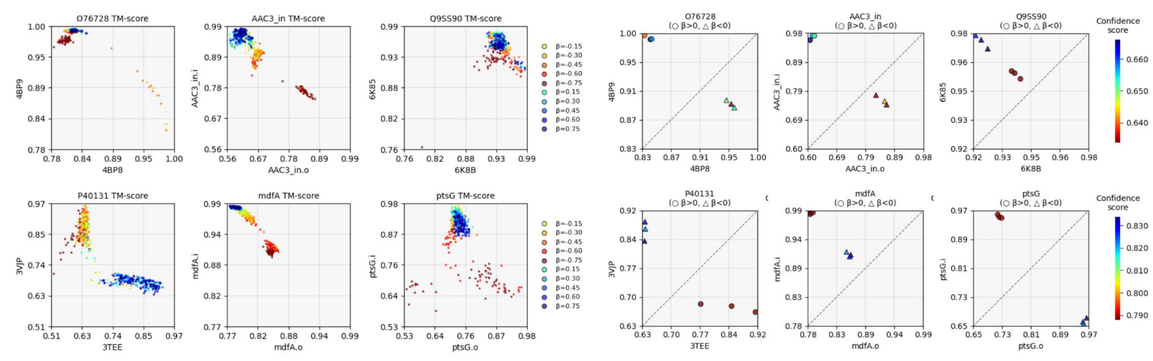

Figure 3. Sampling across several targets. Left panels: TM-scores to the two reference states for ensembles generated with different β values, showing that the sign of β biases the sampled conformation. Right panels: picking only the top 1% high-confidence models yields clear representatives of both states for each target.

Case Study: Hinge Motion in B3EYN2

We applied pair-representation perturbation to B3EYN2, which undergoes a large open-to-closed transition, and found that simply varying β yields accurate models of both the open and closed conformations; the resulting per-residue root-mean-square fluctuation (RMSF) closely tracks the experimental C_α displacement, indicating that the sampled ensemble captures the dominant hinge motion.

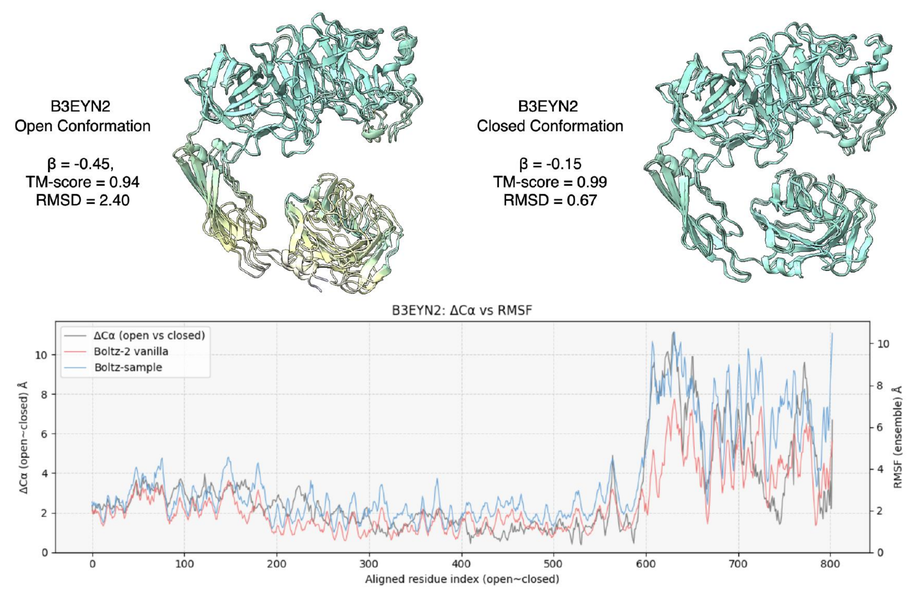

Figure 4. B3EYN2 hinge motion. Open and closed conformations sampled by perturbing the pair representation, shown superimposed on the experimental structures, closely match both states, and the per-residue RMSF of the samples agrees with the experimental C_α displacement.

Case Study: Fold-Switching Proteins

We next applied pair-representation perturbation to proteins that undergo complete fold switching. The method successfully samples both folds for KaiB and RfaH, but still does not reliably recover the open and closed states of MAD2, indicating that some classic fold-switching cases remain challenging.

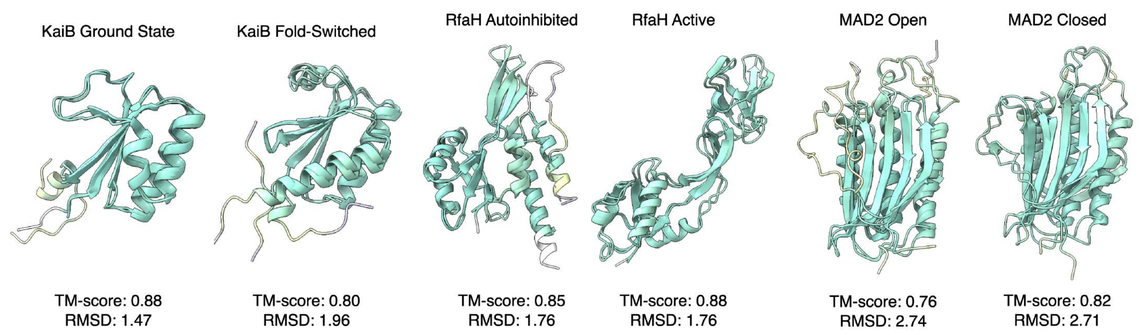

Figure 5. Fold-switching proteins. Pair-representation perturbation recovers both folds of KaiB and RfaH, but not the open and closed conformations of MAD2.

Conclusions

We propose a minimal, interpretable approach to sampling alternative protein conformations by perturbing pair representations in a Boltz-2 model. In this framework, a single parameter β modulates the learned sequence—structure couplings, efficiently reweighting MSA-derived conformational modes without retraining.